

Md Rahatul Ashakin

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Research Focus

Reliable AI through optimization and privacy. I use hybrid classical and quantum methods such as QAOA in Qiskit to improve model selection and robustness, and apply privacy preserving techniques for secure data sharing in health and finance across cloud and edge systems.

Education

M.S., Information Technology (Track: Data Management & Analytics) *Oct 2022–Mar 2024*
GPA: 4.00/4.00, Washington University of Science and Technology, Alexandria, Virginia

B.S., Computer Science and Engineering, North South University, Dhaka, Bangladesh *Sep 2013–Aug 2018*

Publications & Scholarly Contributions (Selected)

Book Chapter (forthcoming)

- **Ashakin, M. R.** (2026). Chapter 5: *Current Tools and Techniques of Artificial Intelligence in Healthcare*. In D. Roosan (Ed.), *Quantum Computing in Medicine: Transforming Drug Discovery, Protein Folding, and Genomics Research through Artificial Intelligence Integration*. Elsevier. ISBN: 978-0-443-34193-9. *CRedit*: Conceptualization; Methodology; Writing: original draft; Visualization; Writing: review & editing.

Refereed Conference Papers

- Roosan, D., **Ashakin, M. R.**, Khan, R., Hasiba Mashed Khan, Khou, T., Carnasciali, M. I., Haider, M. R. (2025). *Variational Quantum Circuits for Molecular Classification Using Graph Neural Network*. In: *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC 2025)*. *CRedit*: Methodology; Software; Formal analysis; Visualization; Writing: review & editing.
- Roosan, D., **Ashakin, M. R.**, Khou, T. (2025). *Adaptive Multimodal Artificial Intelligence with Liquid Neural Network for Edge Computing-Based Augmented Reality*. In: *IEAI 2025, IOS Press, Advances in Transdisciplinary Engineering*. *CRedit*: Conceptualization; Methodology; Software; Visualization; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T., Nirzhor, S., Haider, M. R. (2025). *Quantum Variational Transformer Model for Enhanced Cancer Classification*. In: *IEAI 2025, IOS Press, Advances in Transdisciplinary Engineering*. *CRedit*: Methodology; Software; Validation; Formal analysis; Writing: review & editing.
- Roosan, D., Khou, T., Pham, B., Li, Y., **Ashakin, M. R.**, Hasiba Mashed Khan, Khan, R., Haider, M. R. (to appear, Dec 2025). In: *Quantum AI Enhanced Blockchain Security for Drug Discovery*. *Proceedings of ICIAI 2025, Springer LNEE (to appear)*. *CRedit*: Conceptualization; Methodology; Writing: review & editing.

Under Review (Journal)

- **Ashakin, M. R.**, Stuetzle, C., Khan, R., Nirzhor, S., Roosan, D. *Quantum Annealing of Financial Sentiment Analysis by Optimizing Large Language Model Hyperparameters*. Under review at *Financial Innovation (SpringerOpen)*. *CRedit*: Lead: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing: original draft. Supporting: Data curation; Writing: review & editing. *Resources*: Code: [GitHub repository](#) (Manuscript available upon request).

Academic Service

Research Programmer, Merrimack College (Prof. Don Roosan), Remote (Queens–NY)

- **DSE 4900 (Capstone)** – Fall 2024; Fall 2025: Mentored teams from proposal to final presentation; coordinated labs and weekly clinics; supervised projects/theses end-to-end (scoping, curation, modeling, reporting); instituted reproducible practice (clean repos, env pinning, MLflow) and rigorous evaluation (ablations, error analysis, calibration, leakage/imbalance).
- **DSE 4116/6116 Quantum Machine Learning** – Fall 2025: Designed and delivered Qiskit labs (VQE, QAOA, QSVM/QNN) with classical baselines; mentored hybrid classical–quantum workflows and parameter sweeps; enforced reproducibility (clean repos, env pinning, MLflow) and fair-comparison protocols.
- **DSE 3010 (ML for DS)** – Fall 2024: Taught model selection and diagnostics; standardized repos/environments and MLflow tracking with diagnostic checks (ablations, error analysis, leakage/imbalance).

Professional Experience

Data Science Intern, Tekurai Inc., San Antonio, Texas (remote, Queens - NY) *May 2024–Present*

- Built transformer and CNN pipelines for clinical text and audio, analyzed clinician and patient interactions for decision support.

- Processed 500k+ healthcare records with Python and produced interactive Power BI dashboards for operations and care insights.
- Prototyped **privacy-aware** blockchain workflows for secure data exchange in drug discovery across multiple institutions.
- Quantum machine learning with Qiskit, including hybrid variational models and **quantum-guided** hyperparameter search that improved internal accuracy and AUC over classical baselines.

Program Manager, ShopUp, Dhaka, Bangladesh

Feb 2022–Sep 2022

- Led analytics for B2C and B2B operations with forecasting and KPI reporting; delivered Tableau and Power BI dashboards used to reduce returns and improve throughput.
- Partnered with operations, product, and engineering to define a KPI taxonomy and the supporting SQL/ETL pipelines; automated weekly executive dashboards and cohort analyses to speed up decision cycles.

Project Manager, Icon Information Systems Ltd., Dhaka, Bangladesh

Jun 2020–Feb 2022

- Directed cross-team reporting and SQL data workflows for enterprise clients; coordinated agile ceremonies and risk tracking.
- Produced project plans and stakeholder updates, organized sprints and stand-ups, and prepared UAT and release checklists to ensure on-time, auditable deliveries.

Assistant Project Manager, Millennium Information Solution Ltd., Dhaka, Bangladesh

Nov 2018–May 2020

- Supported delivery of HRIS and financial systems with user acceptance testing, reliability checks, and transaction safety reviews for regulated stakeholders.

Skills

Trustworthy ML & Stats: Transformers · GNNs · multimodal fusion · robust training · calibration & uncertainty · ablations/error analysis

Quantum & Optimization: Qiskit · VQE/QAOA · hybrid C–Q workflows · QUBO/Ising mappings · quantum-guided HPO

Data Eng & Reproducibility: SQL · ETL · schema design · MLflow/W&B · env pinning · unit tests · CI/CD (Actions) · DVC

Visualization & Writing: Matplotlib · dashboards · LaTeX/Overleaf · figure/table design

Privacy & Cryptography: HE (CKKS/BGV; OpenFHE/SEAL) · differential privacy · secure aggregation · PQC (KEMs/signatures; liboqs) · threat modeling

Clinical NLP & Speech: HF fine-tuning · sequence labeling · ASR preprocessing · sentiment · topic modeling

Programming & Platforms: Python (PyTorch, scikit-learn, NumPy, Pandas) · Git/Linux · Jupyter · Docker (basic) · GPUs · cloud · edge/IoT

Domains: Computational genomics · clinical decision support · financial sentiment

Selected Research Projects (open-source)

- **IBD-Quantum-ESM.** Quantum-informed *IBD* modeling pipeline (classical+quantum optimization; Qiskit QSVC/QuantumKernel + ESM mapping). *Resources:* Links: [Webpage](#) · [GitHub](#) · [Source/CLI](#) · **Key Results & Reproducibility:** IBDMDB—F1_{macro} 0.797; ROC–AUC 0.927 (5-fold CV); CLI: ingest-omics → train-omics → report.
- **Quantum Optimization Sandbox.** Scaffold to *repurpose* ML experiments with QAOA/VQE vs. classical baselines. *Resources:* Links: [GitHub](#) · [README](#) · [Notebooks](#) · [Scripts](#) · **Key Results & Reproducibility:** PDB 4DVF docking top-10—best donepezil −8.2 kcal/mol; reproducible via `scripts/` & `envs/wsl.adquant.q.yml`.
- **FinBERT_Quantum_Optimization.** FinBERT fine-tuning with quantum optimization (QAOA/VQE) for financial sentiment. *Resources:* Links: [GitHub](#) · [README](#) · [Notebooks](#) · [Environment](#) · **Key Results & Reproducibility:** Accuracy +11 points (78% → 89%) vs. classical FinBERT; quick start: `git clone`; `pip install -r requirements.txt`; run `notebooks/finetune.ipynb`, `notebooks/QBert.ipynb`.

Online Courses & Certifications (selected)

[Microsoft Power BI Data Analyst Specialization](#) · Microsoft
[Google Business Intelligence](#) · Google
[SAS Statistical Business Analyst](#) · SAS

[Google Advanced Data Analytics](#) · Google
[Google Data Analytics](#) · Google
[Project Management & Other Tools for Career Development](#)
 · UC Irvine